Objective

To implement and demonstrate Interface Inheritance in Java, where interface I1 extends interface I2, and class C1 implements I1.

Steps for Execution

1.Create a Java project and package named javacore.

2.Create a file named Main.java and enter the given code.

3.Define interface I2 with the add() method.

4.Define interface I1 extending I2 with the subtract() method.

5.Create class C1 implementing I1 and define both methods.

6.Create a C1 object and call the methods.

7.Compile and run the program.

8.Observe the output.

Algorithm

1.Start the program.

2.Define interface I2 with add().

3.Define interface I1 extending I2 with subtract().

4.Create class C1 implementing I1.

5.Implement add() and subtract().

6.Create a C1 object.

7.Call both methods.

8.Display the output.

9.Stop the program.

UML CLASS DIAGRAM:

```
┌──────────────────────────┐
   │  <<interface>>   │
   │      I2        │
```

```
├──────────────────────────┤
│ + add() : void     │
└────────────▲────────────┘
        │
        │ extends
        │
┌────────────┴────────────┐
│  <<interface>>    │
│      I1        │
├──────────────────────────┤
│ + subtract() : void │
└────────────▲────────────┘
        │
        │ implements
        │
┌────────────┴────────────┐
│      C1        │
├──────────────────────────┤
│ - a : int = 20     │
│ - b : int = 10     │
│ + add() : void     │
│ + subtract() : void │
└─────────────────────────┘
```

Relationships:

- I1 extends I2
- C1 implements I1
- C1 implements both add() and subtract().

CODE:

```
package javacore;
interface I2 {
   void add();
}

interface I1 extends I2 {
   void subtract();
}

class C1 implements I1 {

   int a = 20;
   int b = 10;
  @Override
   public void add() {
      System.out.println("Addition = " + (a + b));
   }
 @Override
   public void subtract() {
      System.out.println("Subtraction = " + (a - b));
   }
}

public class Main {
   public static void main(String[] args) {

      C1 obj = new C1();
```

```
        obj.add();
        obj.subtract();
    }
}
```

OUTPUT:

Addition = 30

Subtraction = 10

Result:

The program successfully demonstrates Interface Inheritance in Java. Interface I1 extends I2, and class C1 implements I1, thereby implementing both add() and subtract() methods.